Eco-Risk Alpha: ESG Information in CDS Spreads and Counterparty CVA

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Team case study provided by Deloitte

June 21, 2026

1 Introduction

The growing emphasis on Environmental, Social and Governance (ESG) criteria raises a natural question for credit markets: do firms with stronger ESG performance carry lower credit risk? If so, this should be visible in their credit default swap (CDS) spreads and, for banks trading over-the-counter derivatives with them, in a lower Credit Valuation Adjustment (CVA).

This report tests whether ESG characteristics provide incremental predictive information for 5-year CDS spreads beyond traditional financial fundamentals, using a synthetic but realistic dataset of about 60 firms. Any predictive relationship could motivate future causal research, but the analysis itself is not designed to estimate causal effects. Observed and model-implied spreads are then mapped into CVA on a benchmark 5-year payer interest rate swap (IRS), to assess how the models' prediction errors carry over to counterparty credit-risk measures.

The analysis addresses two questions:

1. What is the marginal contribution of ESG variables to CDS spreads, and is it statistically and economically significant?
2. How accurately do the models predict CDS spreads, and how do their prediction errors translate into CVA?

2 Data and Methodology

2.1 Data and preprocessing

The dataset contains about 60 anonymised firms with sector and region classifications, an aggregate ESG score and its E, S and G pillars, carbon intensity, financial fundamentals (market capitalisation, total assets, debt-to-equity, EBITDA margin, leverage ratio) and observed 5-year CDS spreads. For the CVA computation, an Expected Exposure (EE) profile of the IRS and a EUR zero-rate curve are provided, with a fixed Loss-Given-Default (LGD) of 60%.

Preprocessing consisted of four steps. Categorical labels were harmonised (e.g. "Tech" $\rightarrow$ "Technology"). Six missing ESG scores were reconstructed from the pillars as $\text{ESG} = 0.5E + 0.3S + 0.2G$, missing values in carbon intensity and market capitalisation were imputed with the sector median, and one firm with too many missing fields was removed. Duplicate records were

dropped. Finally, market capitalisation and carbon intensity showed a few extreme but plausible values, so they were retained and transformed with $\log(1 + x)$ to reduce skewness. About 50 firms remain for modelling.

2.2 Models

Three models are estimated for the 5-year CDS spread:

- **OLS** on ESG_Score, Total_Assets, Leverage_Ratio and EBITDA_Margin (a parsimonious set chosen to limit collinearity), with White's heteroskedasticity-robust (HC0) standard errors, as an interpretable benchmark;
- **Random Forest** (400 trees, maximum depth 3, at least 6 observations per leaf) and **Gradient Boosting** (600 trees, learning rate 0.01, maximum depth 3), to capture non-linearities and interactions. Both use ESG_Score, log_Carbon_Intensity, Leverage_Ratio, log_Market_Cap, EBITDA_Margin and one-hot sector and region dummies.

All models are evaluated with 5-fold cross-validation, and out-of-sample predictions for each firm are obtained with `cross_val_predict`.

2.3 From CDS spreads to CVA

Unilateral CVA is the discounted expected loss from counterparty default over the life of the swap:

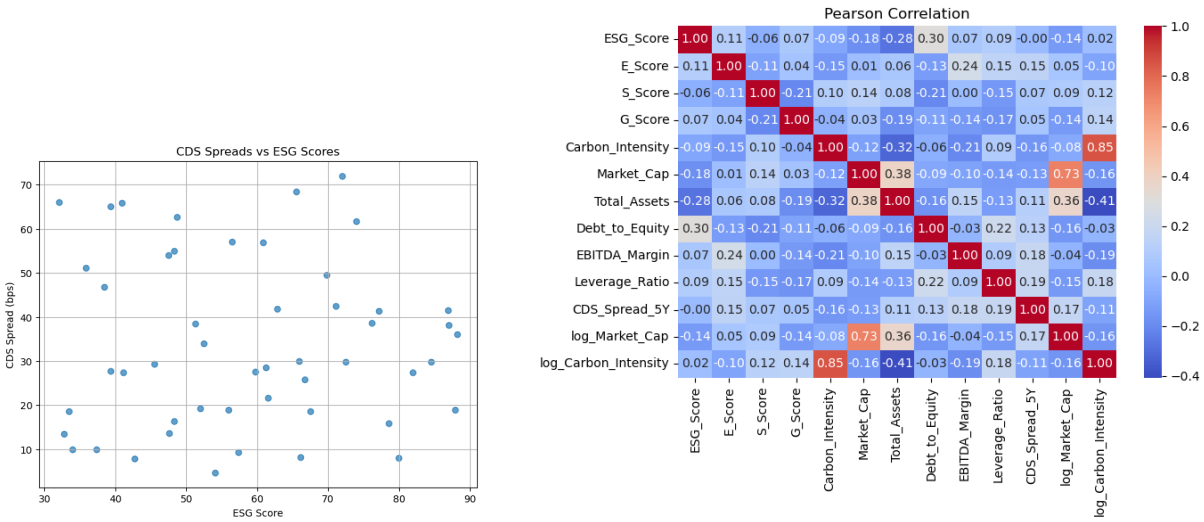
$$\mathrm{CVA} = (1 - R) \int_0^T EE(t)\, DF(t)\, \lambda(t)\, S(t)\, dt \;\approx\; \mathrm{LGD} \sum_{i=1}^{N} EE_i\, DF_i\, (S_{i-1} - S_i)\,,$$

where $R = 1 - \mathrm{LGD}$ is the recovery rate. Each CDS spread s is mapped into a constant hazard rate $\lambda = s/(1 - R)$ and survival probabilities $S_i = \exp(-\lambda t_i)$ on a semi-annual grid. Discount factors $DF_i = e^{-r_i t_i}$ come from the zero-rate curve, linearly interpolated. This yields, for each firm, a CVA based on observed spreads and one for each model's predicted spreads.

3 Results

3.1 Exploratory analysis

The scatter plot of ESG scores against CDS spreads (Figure 1) shows no visible linear or non-linear relationship. Correlations confirm this: the ESG score is essentially uncorrelated with CDS spreads (Pearson ≈ 0, Spearman ≈ 0.04), while EBITDA margin, leverage and log market capitalisation show stronger, though still weak, associations. Pearson and Spearman matrices display very similar patterns.



(a) ESG score vs. CDS spread

(b) Pearson correlation matrix

Figure 1: Exploratory analysis.

3.2 Linear benchmark

The OLS regression confirms the absence of a meaningful linear relationship (Table 1). None of the regressors is significant at the 5% level, the ESG coefficient is practically zero ($p = 0.98$), and the cross-validated R^2 is negative: out of sample, the model predicts worse than the sample mean.

Variable	Coefficient	p-value
ESG_Score	0.0034	0.983
Total_Assets	4.1×10^{-5}	0.496
Leverage_Ratio	5.886	0.112
EBITDA_Margin	0.250	0.249
In-sample R^2 (adjusted)	0.075 (-0.007)	
Cross-validated R^2	-0.123 ± 0.072	
Cross-validated RMSE	19.61 ± 1.64	

Table 1: OLS regression with heteroskedasticity-robust standard errors.

3.3 Machine learning models

The tree-based models do not improve predictive accuracy (Table 2): both have negative cross-validated R^2, with Gradient Boosting performing worse and less stably than Random Forest, a sign of overfitting on such a small sample.

Model	CV R^2	CV RMSE	CV MAE
Random Forest	-0.073 ± 0.161	19.04 ± 1.14	15.63 ± 0.56
Gradient Boosting	-0.325 ± 0.420	20.77 ± 2.18	16.49 ± 2.05

Table 2: Cross-validated performance (5 folds) of the tree-based models.

Feature importances are nevertheless informative about which variables the models rely on

(Table 3). In both models, firm fundamentals dominate, led by log market capitalisation and leverage, while the ESG score ranks fifth. Sector and region dummies contribute almost nothing.

Feature	Random Forest	Gradient Boosting
log_Market_Cap	0.382	0.372
Leverage_Ratio	0.205	0.291
EBITDA_Margin	0.177	0.122
log_Carbon_Intensity	0.103	0.132
ESG_Score	0.094	0.032
Sector and region dummies (total)	0.040	0.051

Table 3: Gini-based feature importances.

Partial dependence and individual conditional expectation (ICE) plots make the contrast clear (Figure 2): predicted spreads change markedly with market capitalisation, whereas the effect of the ESG score is essentially flat. The Gradient Boosting plots show the same pattern.

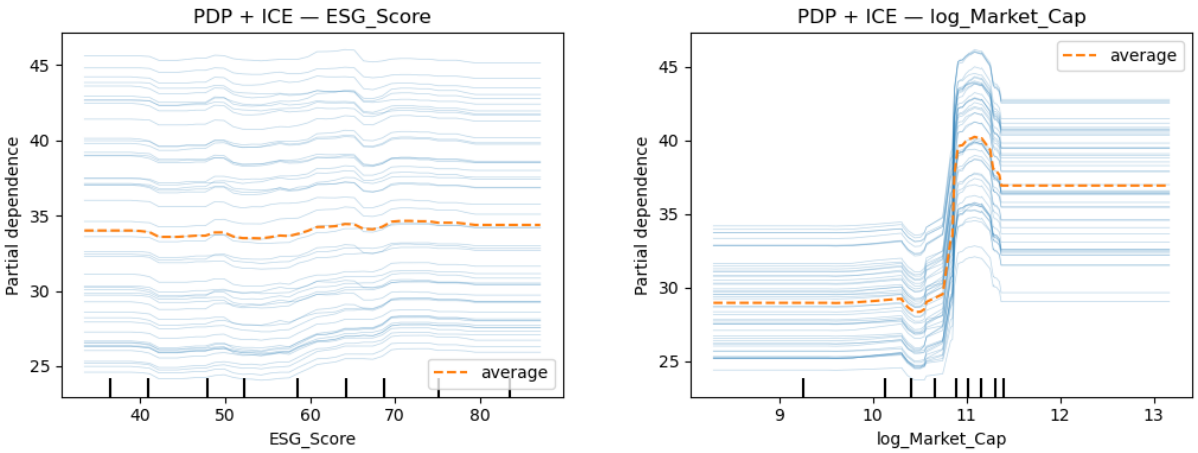


Figure 2: Partial dependence and ICE plots for the Random Forest: ESG score (left) and log market capitalisation (right).

3.4 CVA comparison

To assess model accuracy in CVA terms, the difference between the CVA computed from observed spreads and the CVA computed from model-implied spreads is evaluated for each firm:

$$\Delta\text{CVA} = \text{CVA}_{\text{actual}} - \text{CVA}_{\text{predicted}}.$$

For an accurate model, this distribution should be centred around zero with limited dispersion. Since the predicted spreads depend on both ESG variables and fundamentals, ΔCVA measures the overall prediction error of the models, not the effect of ESG alone.

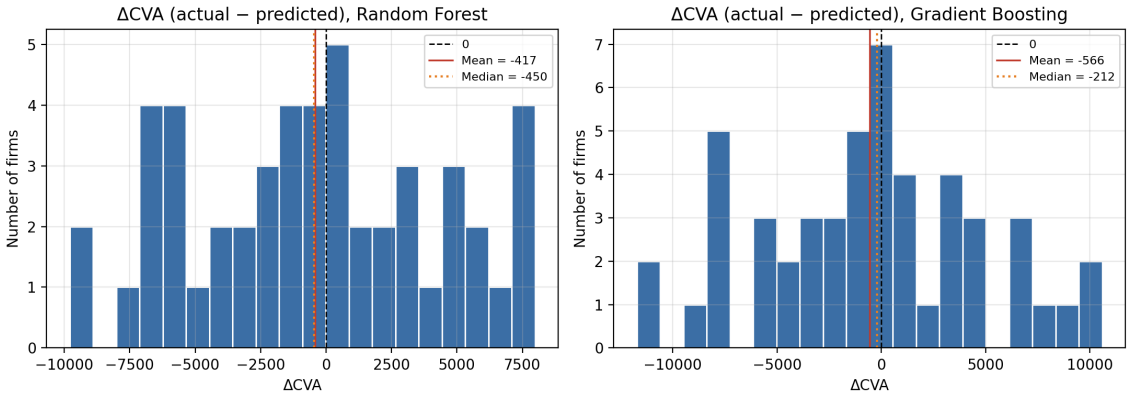


Figure 3: Distribution of ΔCVA across firms for the Random Forest and Gradient Boosting models.

Both distributions are roughly centred around zero (Figure 3): the means are −417 for Random Forest and −566 for Gradient Boosting, so the models do not systematically over- or underestimate CVA. Dispersion, however, is wide, with firm-level differences of about ±10,000. Normalising ΔCVA by the average actual CVA, values range from −1.14 to 0.93 for Random Forest and from −1.37 to 1.23 for Gradient Boosting: for individual firms, the error can be as large as the average CVA itself. This is consistent with the negative cross-validated R^2 and shows that, on this sample, the models cannot be relied on to price counterparty risk at firm level.

A direct test of the contribution of ESG to CVA would compare the CVA implied by models estimated with and without ESG variables. Given the low importance and flat partial dependence of the ESG score, this difference would be expected to be small.

4 Conclusion

This study investigated whether ESG characteristics provide incremental predictive information for CDS spreads beyond traditional financial fundamentals, and how model-implied spreads translate into counterparty CVA.

Across linear and non-linear models, firm fundamentals, especially size and leverage, consistently emerged as the main drivers of CDS spreads, while the ESG score showed a weak and statistically insignificant effect. Overall predictive power was limited, with negative cross-validated R^2 for all models. Mapped into CVA, the model errors are centred around zero on average but large at firm level.

The main limitation is the data: a small, synthetic cross-section constrains statistical power and may hide subtle ESG effects. A weak ESG signal in this sample therefore does not rule out a causal effect of ESG on credit risk; testing it would require richer, real-world data, ideally with a time dimension, and a dedicated research design.